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**Algorithm 5** Evolved survival: replace an old, costly member

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**Input** Population  $P$  with birth indices  $b_i$  and costs  $c_i$ ; excluded replacement indices  $E$ . At least one index is eligible.

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1  $I \leftarrow \{1, \dots, |P|\} \setminus E$ 
2  $b_- \leftarrow \min_{i \in I} b_i$ ;  $b_+ \leftarrow \max_{i \in I} b_i$ 
3  $c_- \leftarrow \min_{i \in I} c_i$ ;  $c_+ \leftarrow \max_{i \in I} c_i$ 
4 for  $i \in I$  do
5    $a_i \leftarrow \begin{cases} (b_+ - b_i)/(b_+ - b_-), & b_+ \neq b_-, \\ 1, & \text{otherwise} \end{cases}$ 
6    $q_i \leftarrow \begin{cases} (c_i - c_-)/(c_+ - c_-), & c_+ \neq c_-, \\ 0, & \text{otherwise} \end{cases}$ 
7    $v_i \leftarrow 0.75 a_i + 0.25 q_i$ 
8 return the first index in  $\arg \max_{i \in I} v_i$ 
```

Smaller birth indices denote older members. Both age and cost are normalized over the eligible population: old age and high cost increase replacement priority. This is a weighted tradeoff, rather than a strict age ordering with a cost tie-break. The returned index is replaced by PySR; the operator does not insert the child.

The equations describe the source's raw min-max arithmetic. It has no special handling for nonfinite costs.

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